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Register Number

Class : 12

COMMOM HALF YEARLY EXAMINATION, 2025 - 26

MATHEMATICS

Time Allowed : 3.00 Hours] [Max. Marks : 90

PART - I 20 x 1 = 20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. If A, B and C are invertible matrices of some order, then which one of the following is not true?

(1) adj(AB) = (adj A)(adj B) (2) adj A = |A| A^{-1} (3) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ (4) $A^{-1} = (\det A)^{-1}$

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2. The principal argument of the complex number $\frac{(1+i\sqrt{3})^2}{4i(1-i\sqrt{3})}$ is

(1) $\frac{5\pi}{6}$ (2) $\frac{\pi}{2}$ (3) $\frac{2\pi}{3}$ (4) $\frac{\pi}{6}$

3. The number of positive zeros of the polynomial $\sum_{j=0}^{n} {}^nC_r(-1)^r x^r$ is

(1) < n (2) r (3) 0 (4) n

4. If $\cot^{-1}(\sqrt{\sin\alpha}) + \tan^{-1}(\sqrt{\sin\alpha}) = u$, then cos 2u is equal to

(1) 0 (2) $\tan^2\alpha$ (3) $\tan 2\alpha$ (4) -1

5. If x + y = k is a normal to the parabola $y^2 = 12x$, then the value of k is

(1) 1 (2) -1 (3) 9 (4) 3

6. The angle between the line $\vec{r} = (\hat{i} + 2\hat{j} - 3\hat{k}) + t(2\hat{i} + \hat{j} - 2\hat{k})$ and the plane $\vec{r}.(\hat{i} + \hat{j}) + 4 = 0$ is

(1) 45° (2) 90° (3) 0° (4) 30°

7. The function $\sin^4 x + \cos^4 x$ is increasing in the interval

(1) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (2) $\left[0, \frac{\pi}{4}\right]$ (3) $\left[\frac{5\pi}{8}, \frac{3\pi}{4}\right]$ (4) $\left[\frac{\pi}{2}, \frac{5\pi}{8}\right]$

8. If f(x, y, z) = xy + yz + zx, then $f_x - f_z$ is equal to

(1) y - z (2) z - x (3) y - x (4) x - z

9. If $f(x) = \int_0^x t\cos t\, dt$, then $\frac{df}{dx} =$

(1) x cos x (2) x sin x (3) cos x - x sin x (4) sin x + x cos x

10. $\int_0^{2a} f(x)dx =$

(1) 0 (2) $2\int_0^a f(x)dx$ (3) a (4) $\int_0^a f(x)dx + \int_0^a f(2a - x)dx$

11. The differential equation of the family of curves $y = Ae^x + Be^{-x}$, where A and B are arbitrary constants is

(1) $\frac{d^2y}{dx^2} - y = 0$ (2) $\frac{d^2y}{dx^2} + y = 0$ (3) $\frac{dy}{dx} - y = 0$ (4) $\frac{dy}{dx} + y = 0$

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12. The number of arbitrary constants in the general solutions of order n and n +1are respectively

(1) n+1, n (2) n+1, n+2 (3) n, n+1 (4) n−1, n

13. If $x^a y^b = e^m$, $x^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ then the values of x and y are respectively,

(1) $e^{(\Delta 1/\Delta 3)}, e^{(\Delta 2/\Delta 3)}$ (2) $e^{(\Delta 2/\Delta 1)}, e^{(\Delta 3/\Delta 1)}$ (3) $\log(\Delta 1/\Delta 3), \log(\Delta 2/\Delta 3)$ (4) $\log(\Delta 2/\Delta 1), \log(\Delta 3/\Delta 1)$

14. If $x = \frac{1}{5}$, the value of $\cos(\cos^{-1} x + 2\sin^{-1} x)$ is

(1) $\sqrt{\frac{24}{25}}$ (2) $-\sqrt{\frac{24}{25}}$ (3) $-\frac{1}{5}$ (4) $\frac{1}{5}$

15. The point of inflection of the curve $y = (x-1)^3$ is

(1) (1,0) (2) (0,0) (3) (1,1) (4) (0,1)

16. A pair of dice numbered 1, 2, 3, 4, 5, 6 of a six-sided die and 1, 2, 3, 4 of a four-sided die is rolled and the sum is determined. Let the random variable X denote this sum. Then the number of elements in the inverse image of 7 is

(1) 4 (2) 3 (3) 2 (4) 1

17. If X~B(n, p) then

(1) $\mu = np$; $\sigma^2 = np(1-p)$ (2) $\mu = nq$; $\sigma = np(1-p)$

(3) $\mu = np$; $\sigma = np(1-p)$ (4) $\mu = npq$; $\sigma = npq$

18. In the last column of the truth table for $\neg(p \vee \neg q)$ the number of final outcomes of the truth value 'F' are

(1) 4 (2) 1 (3) 3 (4) 2

19. If $p + \sqrt{q}$ and $-i\sqrt{q}$ are the roots of a polynomial equation then the least possible degree of the equation is

(1) 2 (2) 1 (3) 3 (4) 4

20. For the parabola $(x-h)^2 = -4a(y-k)$ the equation of directrix is

(1) y = k (2) y = a (3) x = k + a (4) y = k + a

PART - II

1. **Answer any 7 questions**
2. **Each question carries 2 marks**
3. **Question number 30 is compulsory**

7 x 2 = 14

21. Find the adjoint of $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$

22. Simplify : $i^{-1924} + i^{2018}$

23. Evaluate : $\int_3^4 \frac{dx}{x^2-4}$

24. Show that the $y = ae^x + be^{-x}$ is a solution of the differential equation : $y'' - y = 0$

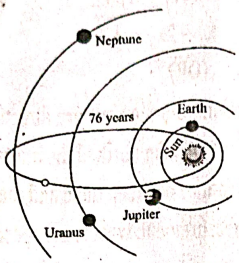
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25. Write Maclaurin's series for e^{-3x}.

26. A pair of fair dice is rolled once. Find the probability mass function to get the number of fours.

27. The orbit of Halley's Comet is an ellipse 36.18 astronomical units long and by 9.12 astronomical units wide. Find its eccentricity.



The elliptical orbit of Halley's Comet

28. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. If $c_1 = 1$ and $c_2 = 2$, find c_3 such that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar

29. Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ be any two boolean matrices of the same type. Find $A \vee B$ and $A \wedge B$.

30. Find the value of $\sin^2\left(\tan^{-1}\frac{3}{4}\right)$

PART - III

1. **Answer any 7 questions** 7 x 3 = 21
2. **Each question carries 3 marks**
3. **Question number 40 is compulsory**

31. Solve the following system of linear equation by matrix inversion method: $2x - y = 8$, $3x + 2y = -2$

32. Find the binomial distribution for each of the following.
(i) Five fair coins are tossed once and X denotes the number of heads.
(ii) A fair die is rolled 10 times and X denotes the number of times 4 appeared.

33. Polar form of $-3\sqrt{2} + 3\sqrt{2}i$

34. If $x^2 + 2(k+2)x + 9k = 0$ has equal roots, find k.

35. Find the value of $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right) + \cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)$

36. A race car driver is kilometer stone 20. If his speed never exceeds 150 km/hr, what is the maximum kilometer he can reach in the next two hours.

37. If $u(x,y) = x^2y + 3xy^4$, $x = e^t$ and $y = \sin t$, find $\frac{du}{dt}$ and evaluate it at $t = 0$.

38. Verify (i) closure property, (ii) commutative property, (iii) associative property for following operation on the given set.
$m * n = m + n - mn$; $n \in Z$

39. Prove that $[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}] = 0$

40. Solve : $e^x\sqrt{1-y^2}dx + \frac{y}{x}dy = 0$

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PART - IV

7 x 5 = 35

i) Answer all the questions

ii) Each question carries 5 marks

41. a) Solve the following system of equations, using matrix inversion method:

$2x_1 + 3x_2 + 3x_3 = 5$, $x_1 - 2x_2 + x_3 = -4$, $3x_1 - x_2 - 2x_3 = 3$.

(OR)

b) A police jeep, approaching an orthogonal intersection from the northern direction, is chasing a speeding car that has turned and moving straight east. When the jeep is 0.6 km north of the intersection and the car is 0.8 km to the east. The police determine with a radar that the distance between them and the car is increasing at 20 km/hr. If the jeep is moving at 60 km/hr at the instant of measurement, what is the speed of the car?

42. a) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$, show that $x^2 + y^2 + 3x - 3y + 2 = 0$.

(OR)

b) Evaluate $\int_0^{\pi} \frac{x}{1+\sin x} dx$

43. a) If the roots of $x^3 + px^2 + qx + r = 0$ are in H.P., prove that $9pqr = 27r^3 + 2p$. Assume $p, q, r \neq 0$.

(OR)

b) Find the population of a city at any time t, given that the rate of increase of population is proportional to the population at that instant and that in a period of 40 years the population increased from 3,00,000 to 4,00,000.

44. a) Find the equation of the circle through the points (1,0), (−1,0) and (0,1).

(OR)

b) A random variable X has the following probability mass function.

x	1	2	3	4	5	6
f(x)	K	2k	6k	5k	6k	10k

Find (i) $P(2 < X < 6)$ (ii) $P(2 \leq X < 5)$ (iii) $P(X \leq 4)$ (iv) $P(3 < X)$

45. a) Find the parametric vector, non-parametric vector and Cartesian form of the equations of the plane passing through the three non-collinear points (3,6,-2), (-1,-2,6), (6,-4,-2).

(OR)

b) Establish the equivalence property connecting the bi-conditional with conditional : $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

46. a) Prove by vector method that $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$.

(OR)

b) We have a 12 unit square piece of thin material and want to make an open box by cutting small squares from the corners of our material and folding the sides up. The question is, which cut produces the box of maximum volume?

47. a) Find the equation of tangents from the point (2, −3) to the hyperbola $2x^2 - 3y^2 = 6$

(OR)

b) Find the approximate value of $\sqrt[3]{1.02} + \sqrt[4]{1.02}$.

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